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INTRODUCTION

1.1 THESIS TIMELINE (WEEKS)

1. Read & Background
2. Read
3. Setup pipeline & get used to metrics
4. Setup pipeline
5. Add speech
6. Add speech
7. Add speech
8. Present
9. Write
10. Write
11. Write
12. Present

1.2 THESIS OVERVIEW

- * Neural networks deliver similar representations of multimodality data
- * Two representations are present: Platonian (fully converge embeddings) and Aristotelian (nearest neighbor similarities)

- * These are on vision and language aka text and images, the aim is to extend it to audio.

Sections to consider:

- * Types of simialrity metrics and their drawbacks and assumptions: depth and width cofounders, orthogonality assumptions etc.
- * Which models to compare and how adequetely to compare them (differnt depth, size, datasets, losses)
- * How to compare text speech audio

Actions: Run pretrained encoders for speech (wave2vec, HuBert) text (LLama, Bloom, openLLaMa) image (ResNet, transformers, ImageNet-21K, MAE, DINOv2, CLIP, CLIP-finetuned) Proposed model for similarity between modalities.[3] Novel network architectures that learn high-level semantic concepts [4]: other models (doesn't have a name but its like VGG.)

Run similarity metrics:CKA, SVCCA, NN, model stiching.- the null hypothesis, mKNN, RSA, procrusteres.

Possible sections:

- * Explian Platonic and Aristotelian
- * Explain the issue with measureing similarity
- * Explore different similarities, why they work and assumptions
- * Explore null hypothesis calibration
- * Setup expriments to measure said simialrity metrics (MVP)
- * Redo some experiments to see that it is correct and reproducible
- * Add on audio (consider which layers and the training of the data)
- * Do small experiemnts
- * Scale up to larger data

Things to think about: loggin and saving data as to not rerun experiemnts What data do we want Which level of embeddings do we want What are we comapring?: Can audio join platonic and or aristotelian? aka does it also have width and depth cofounder? or can it cluster data only?

LITERATURE

2.1 BACKGROUND

Neural Networks are an intelligent representation of data consisting of nodes linked with weights, during training, these weights are updated via backpropagation and they form the basis of information needed to retrieve a certain output from a given input. After being trained on given data, the weights contain information which can be used for downstream tasks, predicting the output of unseen input data. This is classical networks, recently the use of unsupervised neural networks are growing. They learn patterns, structures and lower dimensional representations without being presented with explicit labelled data. For example DINO and CLIP, BERT, GPT, wav2vec 2.0 and HuBert

"Overall, deep networks are thus generally assumed to be powerful and flexible learning machines that are however not well understood theoretically (Bengio, 2009)"

Moving beyond the workings of neural networks, one could ask the question. If neural networks are trained to solve similar problems, do they discover similar internal representations / weights? Multimodal systems are being built for multiple tasks (text and audio processing), models are becoming homogeneous in their tasks and subsequently their internal representations [6].

Hug [6] states a Platonian hypothesis in which different models are all trying to arrive at a representation of reality (a joint distribution over events in the world that generate the data we observe). they consider the convergence of the hypothetical representation of real data in a lower dimension to that agrees between models, the "platonic representation". In reference of Plato's Allegory of the Cave **TODO: cite? and rewrite**, and his idea of an ideal reality that underlies our sensations

This is related to "convergent realism" - that science is converging on truth and closely related to the "Anna Karenina scenario", (Bansal 2021), referring to the possibility that all well-performing neural nets represent the world in the same way.

They found that early layers in networks converge in similarity and converge with increasing size over different architectures, datasets and modalities. Similarity was observed between vision and language models, this all suggests that there is a universal semantic structure.

This view was later contradicted with a Aristotelian view. Groger [2] identifies confounding factors of neural networks (width and depth), which let simialrities between layers appear higher than they actually are. After correcting pervious similarity experiments the previously identified global alignments become much weaker, however a local neighborhood underlying strucutre still remains.

This shifts the debate among mutimodal network representations from whether they converge, to how the convergence should be measured.

Limitations to measuring similarity include: **TODO: find cites** different dimentionalities different neuron orderings different scales

So they are difficult to compare directly, different measurement metrics have been explored to measure the similarity between representations.

Linear Probe

CCA (this is all from [12]) "The purpose of CCA is to find linked sources of variability that describe the characteristic correspondence between two sets of variables, typically capturing two different levels of observation. The relations between each pair of factors across variable sets can be used to compute notions of the conjoined variance explained across both domains" It aims to find useful projections of high dimentional variable sets into compact linear representations. (canonical variates) " Each resulting canonical variate is computed from the weighted sum of every original variable indicated by the canonical vector." CCA finds any linear combination of a set of variables that most highly correlates with any linear combination of another set of variables.

- Symmetry: sets are mutually exchangeable this is its advantage over other types of metrics like linear regression. Linear regression indicates that some change on a dependent variable retrieved a response as a function of the independent variable. So dependent and independent cannot be exchanged to get a simialr result. However in CCA not based on dependent or independent, we can interchange and recieve the same result. - Multiplicity: mode - pair of canonical variates that share variance in both sets

Essentially PCA that maximises Correlation.

Its been widely used in neurology to map brain networks.

CCA is limited to datasets which have more observations than variables. (the width of a network cannot exceed the number of datapoints) Additionally, with increasing number of variables in one or both of the sets, the ensuing canonical correlation often tends to increase, due to higher degrees of freedom. An importance consequence is that canonical correlation obtained from CCA applications with differently sized variables sets cannot be used to decide which the obtained CCA models should be selected as "better". The

CCA solution is constraint by the sample as well as the number of features. It assumes additivity, thus ignoring more complicated variable-to-variable relationships that may exist in the data.

CCA is probably the most general statistical approach to distill the relationships between two high-dimensional sources of quantitative measurements.

An extensions is KCCA first projects the data into this enriched feature space before performing CCA in that enriched input space. It is advantageous that KCCA allows for the detection of complicated non-linear relationships in the data. The drawback is that the interpretation of variable contributions in the original variable space is typically more challenging and in certain cases impossible. Further, KCCA is a nonparametric method; hence, the quality of the model fit scales poorly with the size of the training set

$$p = \frac{\langle w^\top L_1, s^\top L_2 \rangle}{\|w^\top L_1\| \|s^\top L_2\|}$$

CCA was used on neuron activation vectors over a full dataset X. These vectors are the collection of outputs of a single neuron, resulting in one multidimensional variate.

SVCCA: Singular Vector Canonical Correlation Analysis *finding the most important neurons in a network* analysis of each neuron's activation vector. Every neuron in a network does not imply useful information, the aim of SVCCA is to find the directions in the neuron space that carry the most information. SVCCA applied Singular Value Decomposition to layer activations: $X = U\Sigma V^\top$. Then CCA is used to compare the remaining subspaces. These authors [10]. compared the different layers at different points during training and found that it converges bottom-up: the early layers stabilize first- they implemented freeze training to these layers once they stabilize. They also applied Fourier transforms and other experiments, I won't be looking into that for now. *a nice summary of the papers:* <https://maddyschiappa.medium.com/svcca-summary-e83a53f7dd68>

A challenge with this methodology is that the mean correlation coefficient will typically underestimate the degree of similarity. This is because CCA assumes that all c vectors are equally important to the representations in a given layer and it has been supported that deep neural networks do not rely on the full dimensionality of their layers.

PWCCA <https://maddyschiappa.medium.com/summary-pwcca-1909266089e3> is a summary of this paper: <https://arxiv.org/abs/1706.05806> Extension on the observations that networks converge bottom up. Do non-feedforward ones do the same? Because network representations converge early in training, researchers used a weighted mean for those layers that converge early on (and are deemed important). They first determine how much of the original output is accounted for by each CCA vector which is normalized. Finally, they compute the projection weighted CCA distance using the normalized weights α . What they found: They found that generalized networks converged to more similar solutions between each other while the memorizing networks were different from both the generalized networks and each other, **this supports the converging theory of generalized and regularized networks**. "What the researchers found additionally, which I find

very interesting, is that in both networks that memorize r generalize, networks converged to equally similar solutions in early layers. They attribute this to feature detectors being found in early layers of CNNs and later layers converge to more generalizable solutions."

"A new concept called the "lottery hypotheses" hypothesizes that larger networks converge to better solutions because they are more likely to contain a successful sub-network with a "lucky" initialization <https://arxiv.org/abs/1803.03635>. The researchers used PWCCA to investigate whether these larger networks actually contained similar sub-networks. They did this by training groups of CNNs with increasing number of filters at each layer proceeded by computing pairwise similarity between each group of networks of the same size. They found larger networks more often converged to similar solutions with higher accuracy than smaller."

They also found that networks with similar learning rates converge to similar representations, leading to a possible local minima in the dimension space.

They did experiments on non-feedforward networks (RNNs), RNN does exhibit the bottom-up convergence between layers in which the weights are adjusted but is less successful in finding similarity between sequences where weights are not adjusted

RSA

RSM (<https://www.semanticscholar.org/reader/8f80904e8bc53da6af6b7858652d1e65b7c92add>)

CKA a paper: <https://www.semanticscholar.org/paper/0-N-THE-I-NADEQUACY-OF-CKA-AS-A-M-EASURE-OF-S-IN-D-Davari-Horoi/00a41b14598b8a4f0a65f01380cdb8807d0bab71> Instead of just looking at individual pairs, CKA assesses how well the entire structure of one embedding space aligns with another. It essentially asks: "Can we linearly transform one set of embeddings to resemble the other?"

CKA is good for: validating global alignment (ensure the entire embedding space is similar), comparing multi-lingual models, understanding representative learning.

At Scale, the performance is highly dependent on the measurement

"In this paper, we take a closer look at the experimental evidence for the hypothesis and find it to be fragile and to depend critically on the evaluation regime"

<https://www.semanticscholar.org/reader/5b9d3d8b70ffb5e6ca9a438dc93c1de314619242>

They make some concerning conclusions:

They argue that nearest neighbor becomes very strict with larger models (image-text cluster queries). In this paper, we examine how cross-modal alignment changes in evaluation settings with large, dense, and non-bijective datasets, and observe the following: (one-to-one dataset) They found that at an increased scale, the cross-modal nearest neighbors decrease. In controlled settings (e.g., ImageNet), vision and language models reliably

retrieve correct-class neighbors but rarely agree on the same instance. many-to-many correspondence reduces alignment The claim that stronger language models align better with vision seems to weaken for more recent models.

They did some experiments with the threshold of k (the number of clusters/ items in the cluster): interestingly, alignment at $k=n/100$ remains relatively stable across scales, suggesting that models share some degree of coarse structural agreement.

They observe: Observation 1: Mutual kNN alignment scores decrease for denser galleries for fixed small k , suggesting that mutual kNN is sensitive to gallery sparsity. Furthermore, the reported trend that stronger language models align better with vision weakens substantially for fixed small k at WIT-1M scale, with all models scoring near zero

A bad kNN between models means they either fetch the wrong image, or the right one but different images. They investigate this and observe: Observation 2: As the data gets denser, both models retrieve correct-class neighbors at increasing rates, yet strict cross-modal mutual kNN alignment is flat at 11%. This is not a failure of unimodal representation quality, but of unaligned representational organization across modalities.

They observe that you cannot use kNN on data that is not one-to-one.

They test text-audio and text-video to test their findings: Our findings again suggest that models independently trained on different modalities organize information differently. Across text-image, text-video, and text-audio pairs, cross-modal alignment is limited (at fine granularity), consistent with the broader picture of modality-specific organizational structure rather than convergence to a shared representation

The goal was to see whether recent, stronger LLMs continue the alignment trend predicted by Huh et al. The answer was no. This is due to the older benchmarks of model relying on pretraining. They also found: newer models do not become increasingly aligned with vision representations, the scaling trend predicted by Huh et al. breaks down, some of the strongest models are less aligned than expected.

They hypothesize that: Model intelligence can continue increasing without representations becoming more similar across modalities. This supports an Aristotelian interpretation: different modalities organize information in compatible but distinct ways.

They put some emphasis on future work looking at which components correspond within text and image models. The authors therefore argue that future research should identify this shared intersection between modalities, rather than assuming complete representational equivalence.

However, these analyses are restricted primarily to vision and language models. If a modality-independent representational structure truly exists, it should also emerge in self-supervised speech models. Conversely, if speech representations differ substantially, this would provide evidence against a universal Platonic representation.

Related research / other metrics? / contra-

dicting theories <https://www.semanticscholar.org/paper/Barycentric-alignment-for-instance-level-comparison-Saha-He/dae3983be689774b6cd344e9e839877c7c61cfe5>

Although CKA became a standard representation similarity metric, more recent work has shown that even CKA can be influenced by architectural confounders, motivating permutation-based baselines and topology-aware comparisons.

Most recent work has focussed on vision-vision, language-language and text-text representation alignment. For example CLIP, ALIGN and SigLIP. Speech has received little attention, especially self-supervised models comparison of audio and visual models and large-scale representative analysis.

<https://www.semanticscholar.org/reader/5b9d3d8b70ffb5e6ca9a438dc93c1de314619242> they propose Umwelt.

2.1.1 THE PLATONIC REPRESENTATION HYPOTHESIS, [6]

The central idea is the networks are converging to similar representations of the data that they learn. We look specifically at the vector embeddings.

"Different models are trying to arrive at a representation of reality" - the joint distribution over events in the world that generate the data we observe.

It is based on several ideas:

- * "platonic representation", "ideal reality that underlies our sensations".
- * "convergent realism" **TODO:** read: (Newton-Smith, 1981; Putnam, 1982; Doppelt, 2007; Hardin & Rosenberg, 1982), science is converging in truth
- * representation learning literature **TODO:** Tian et al. (2020a); Zimmermann et al. (2021); Richens & Everitt (2024); Cao & Yamins (2024)
- * "Anna Karenina scenario" **TODO:** Bansal et al. (2021) - "all well-performing neural nets represent the world in the same way"

The platonic representation hypothesis refers to the situation where we are in an Anna Karenina scenario and the "happy representation" that is converged upon is one that reflects a statistical model of the underlying reality.

They look at vector embeddings in the kernel space and measure the distances between them. They use a combination of **TODO:** Centered Kernel Distance (CKA), SVCCA, nearest-neighbor metrics to get to their own mutual nearest-neighbor metric that measures the mean intersection of the k-nearest neighbor sets induced by two kernels, K1 and K2, normalized by k.

They explore how several neural networks are converging. First in aligned representation and then that it holds across modalities (image in vision models and text in language models).

TODO: Model stitching: Lenc & Vedaldi (2015) , they found: (1) A vision model trained on ImageNet (Rusakovsky et al., 2015) can be aligned with a model trained on Places-365 (Zhou et al., 2017) while maintaining good performance; *model independence: different image datasets lead to similar representations* (2) The early layers of these convolutional networks are more interchangeable than later layers. *oriented* **TODO: Gabor-like filters** are common in both artificial and biological vision systems and convergence of initial layers

Bansal et al. (2021) expanded on the idea of model stitching, showing that models trained using self-supervised objectives align closely with their supervised counterparts. **Self-supervised comes in** Moschella et al. (2022) further demonstrated the feasibility of “zero-shot” model stitching without learning a stitching layer. In particular, they considered the kernel K defined by learned representations and showed that K serves as a bridge between models, Dravid et al. (2023) extended this idea to individual neurons, and found “Rosetta Neurons” that are activated by the same pattern across a range of vision models. Such neurons form a common dictionary independently discovered by all models.

Model stitching therefore started off by seeing that the same model trained on different data has similar embeddings, and the early layers seem to converge. This was extended to showing that self-supervised and supervised had similar representations, and zero shot. A layer K can serve as a bridge between models. And neurons are rosetta neurons activated in the same way across vision models.

They found that larger models tend to have larger overlap. Models with similar output have similar internal structures. (optimal learning) (A monte carlo like convergence as N = model size increases)

they did many experiments and some maths with bijective representation spaces. They compared similarity metrics.

Importantly, these networks have different objectives trained for different functions on different data.

Assumptions: text and image are related. dataset is representative "Nearest-neighbor overlap measures semantic alignment. This is plausible, but nearest-neighbor identity also depends on density, sample size, anisotropy, preprocessing, and the value of k ." Scale is a meaningful axis of convergence. A shared statistical structure is equivalent to movement toward a common representation. This is the largest interpretive leap.

2.1.2 REVISITING THE HYPOTHESIS, [2]

a permutation-based null-calibration framework that transforms any representational similarity metric into a calibrated score with statistical guarantees.

Aristotelian Representation Hypothesis: representations in neural networks are converging to shared local neighborhood relationships.

They identify cofounders that influence similarity metrics:

- * model width: null inflation- the dimension of the space is larger than the input so similarity is higher by chance and not representative of true similarity.
- * model depth: choosing the correct layer is difficult so researchers take all and then take the max. deeper models can appear more similar simply because they provide more chances to obtain a high similarity score

These introduce systematic bias into similarity metrics. comparing raw similarity scores across models of different widths or depths can lead to incorrect conclusions about how similarly the models represent information

They then introduce " null-calibration for representational similarity, a general permutation-based framework that transforms any similarity metric into a calibrated score with a principled null reference, here defined as no relationship"- this measures how extreme a similarity score is relative to an empirical null distribution obtained by breaking sample correspondences.

- * Width cofounder: They choose some threshold under which the score sets to zero for a certain width of a network.
- * Depth cofounder: aggregation-aware calibration, compute the null distribution of the same aggregate statistic that is ultimately reported (e.g., the maximum over all layer pairs), thereby calibrating the selection step itself.

After considering these biases the authors found that local neighborhood-based metrics retain significant cross-modal alignment However, we also observe that the convergence in local distances is not preserved, suggesting that only local neighborhood relationships are aligned. Motivated by these results, we refine the original Platonic Representation Hypothesis and propose the Aristotelian Representation Hypothesis¹ : Neural networks, trained with different objectives on different data and modalities, converge to shared local neighborhood relationships

It is called Aristotelian because: Greek philosopher Aristotle, who was a student of Plato and, in his Categories, established the principles of relatives

A big issue in this field is choosing the correct similarity metric since this can alter the conclusion of the research.

They considered: (i) spectral: metrics defined on the spectrum of cross-covariance or Gram matrices (e.g., CKA, CCA), (ii) neighborhood: metrics measuring local topological overlap (e.g., mKNN), and (iii) geometric: second-order isomorphism metrics (e.g., RSA).

They defined a null hypothesis as baseline: the absence of a relationship between X and Y beyond their marginal statistics

Made assumptions: Exchangeability under the null: if no true relationship exists, the observed pairing is statistically indistinguishable from a random shuffling of the data because it allows us to construct an empirical null distribution by holding X fixed and shuffling the rows of Y.

TODO: They did some maths

Then they did some **experiments**:

- * Does calibration remove the width cofounder: They created two random matrices (independent) which should have no similarity (this is the null hypothesis). They varied the width, similarity was shown to increase with d/n . They calculated popularity matrices. **TODO:** (Linear CKA, RBF CKA, SVCCA, PWCCA) - compare covariance structure. Compared neighborhood metrics: **TODO:** mKNN and compared overall geometry **TODO:** RSA, procrustes. They expect a similarity of zero. They observed a width cofounder. They then apply calibration: observed - expected under independence resulting in zero similarity for all sizes.
- * Does calibration still detect real signals? They created a common signal and added noise, and varied signal strength, as the signal increases, the calibrated similarity also increases, so the calibration removes fake similarity and keeps the real.
- * Depth Cofounder: Multiple networks with random layers (no true relationship). They compute layer similarities. They calibrate the final aggregation statistic (e.g. avg over all layers). If you only calibrate every cell individually, the maximum still increases with depth. Their aggregate-aware calibration removes this effect.
- * Revisit the Platonic Representation Hypothesis, move to real data. Previously they saw: better language models become increasingly similar to vision models. Without calibration their results show convergence for CKA. Their calibration results indicate that larger models do not converge. Local nearest neighbors similarities did agree but the distances did not. And CKNNA stayed similar.
- * they also tested on video

They mention the simplicity bias. Models are often regularized to reduce overfitting aka they become more generalized for a task. This may improve the convergence theory since models tend toward the simplest most generally applicable solution space.

Assumptions Permutation is a valid null. This requires exchangeability of the sample correspondences under the null. The chosen probe samples are independent enough for the intended inference. The permutation null represents “no relationship.” If there are confounders shared by the two modalities—class labels, dataset construction, duplicate images, or metadata—permutation may remove meaningful structure along with nuisance structure. Calibration fixes the important bias. It addresses width, depth, and aggregation, but not probe-set selection, distribution shift, model selection, or dependence among model comparisons. Local-neighborhood agreement is semantically meaningful. It may be, but it can also reflect shared labels, dataset hierarchy, or coarse category structure.

In terms of code: They made a python importable function for the calibration metric they used to mitigate the confounding width and depths of the models (which permutes the information to find the actual similarity CKA). This includes their scalar null calibration algorithm:

Aggregation-aware calibration for layer-wise comparisons. When comparing models with multiple layers and reporting a summary statistic (e.g., maximum similarity across layer pairs), the aggregation step must also be calibrated. Algorithm 2 shows how to extend scalar calibration to this setting. The key insight is that the same sample permutation must be applied consistently across all layers.

Their code is object oriented research software.

Additional paper from a researcher from this paper: They introduce STRUCTURE which keeps geometrical information for unimodal encoders. https://scholar.google.com/citations?view_op=view_citation&hl=zh-CN&user=nRlBJYQAAAAJ&citation_for_view=nRlBJYQAAAAJ:hq0jcs7Dif8C "To achieve this, we introduce STRUCTURE, a regularization term that preserves the neighborhood relationships of the pretrained space of unimodal encoders within the aligned space." - a limitation is larger data and models (like CLIP)

2.1.3 CROSS MODAL REPRESENTATION AT SCALE [7]

project page: https://akoepke.github.io/cave_umwelten/#fn-related code: https://github.com/akoepke/cave_umwelten

What they did: They used the same metric as the original paper and showed that their experimental evidence breaks at scale, with more realistic (non one-to-one) data and newer models. They also list other papers who disagree with the original paper.

Specifically, we take the original experimental setup and show that alignment degrades when scaling from 1,024 to millions of samples, breaks when the data is not bijective,

captures only coarse categorical agreement rather than fine-grained structure, and the predicted trend that stronger models align more fails for newer LLMs. This suggests that rather than finding a shared representation, modalities may yield representations specific to their own Umwelten (the idea that each creature / model sees the world from its own point of view in different representations.)

TODO: Also notice how they mention bijective datasets, in the case of Flickr8k, it is one-to-many (one image, many captions)

Back to what they did... they used a different dataset, and used subsets thereof. They extracted features saved at every transformer layer so any layer pair can be selected at alignment time. (they wrote their own feature extraction code to Extract DINO or OpenLLaMA features and saved them as pytorch tensors.) Image -> DINO -> ViT -> Layer representation (CLS token per layer, ViT) Text -> Ollama lang model -> Layer representation (pooled hidden state per layer) **TODO:** Audio -> Wave2vec -> Layer representaiton

normalize image, ensure RGB, load dataset (streaming = True), check CUDA Streaming avoids downloading and materialising the full one-million-example dataset in memory. They split the captions into captions and gemini created captions. Vision model immediately into eval mode. Extract every transformer block. Stack the batches. For language they set outputhidden layers to true to get all the layers, also to eval mode. They tokenize and pad the text. One potentially important implementation detail is that it tokenises the entire shard at once, not one batch at a time. (not sure why maybe because they need max length?)

The model produces one vector for every token, but the alignment experiment requires one vector for the whole caption. They do this in two ways: average pooling, or last-token pooling.

They then find the best layers, they align and compute mutual-kNN. Their plotting code is available aswell. https://github.com/akoepe/cave_umwelten/blob/main/plot_nested_evolution.py

TODO: they don't share the find best layer code...

However they find the best layers and then compute the alignment metrics. They use the same layers from each model image / caption pair. Build separate FAISS nearest-neighbour indices For every query: find nearest neighbor for DINO find nearest for LLM calculate overlap. Then report the average overlap across all the dataset items (aka images).

They use L2 normalisations: Every feature vector is divided by its Euclidean norm. This means the nearest-neighbour search becomes sensitive mainly to the direction of each representation rather than its magnitude. It is related to cosine similarity which they can convert to by $similarity = 1.0 - distance/2.0$. FAISS's IndexFlatL2 returns squared Euclidean distances, so this conversion produces cosine similarity when the vectors have

unit norm

In summary, FAISS itself is simply the algorithm used to efficiently compute nearest neighbours. The mathematical object is the k-nearest neighbour operator

2.1.4

2.1.5 SIMILARITY OF NNS REVISITED [8]

Is CKA actually a good measure of representation similarity?

They introduce another metric aswell. This similarity index is equivalent to centered kernel alignment (CKA) and is also closely connected to CCA. Unlike CCA, CKA can reliably identify correspondences between representations in networks trained from different initializations. We show that CCA belongs to a family of statistics for measuring multivariate similarity, but that neither CCA nor any other statistic that is invariant to invertible linear transformation can measure meaningful similarities between representations of higher dimension than the number of data points. *they are also aware that higher dimension representations similarity are biased*

Their main question: An effective method for measuring representational similarity could help answer many interesting questions, including: (1) Do deep neural networks with the same architecture trained from different random initializations learn similar representations? (2) Can we establish correspondences between layers of different network architectures? (3) How similar are the representations learned using the same network architecture from different datasets?

They found that, We show that CKA is able to determine the correspondence between the hidden layers of neural networks trained from different random initializations and with different widths, scenarios where previously proposed similarity indexes fail and verify that wider networks learn more similar representations, and show that the similarity of early layers saturates at fewer channels than later layers. We demonstrate that early layers, but not later layers, learn similar representations on different datasets. **TODO: is this because early layers are larger and later ones smaller?**

They consider that similarity should be invariant to orthogonal transformation and isotropic scaling, but not invertible linear transformation. **invariant to invertible linear transformations:** a limitation of assuming this is that any invariant similarity index gives the same result for any representation of width greater than or equal to the dataset size. They continue to prove this. CNNs for example have more neurons in some layers than there are datapoints in the train set. And training of the network is not invariant to invertible linear transformation of inputs or activations (Gradient descent for example sees rotated data as new datapoints).

A similarity index that is invariant to invertible linear transformation ignores this aspect of the representation, and assigns the same score to networks that match only in

large principal components or networks that match only in small principal components.

invariant to orthogonal transformations: a weaker requirement than the previous. for full rank matrices. it implies invariance to permutations. orthogonal: rotate, reflect and permutation of coordinates.(geometry preserving) This condition bypasses the issue of $d > n$. Orthogonal transformations preserve scalar products and Euclidean distances. If you have a similarity metric between X and Y , to make it invariant to invertible transformations you remove the scaling factor. This can be done by QR decomposition (Q spans the subspace of X) to instead compare orthogonal basis instead of orthogonal representations.

Invariance to Isotropic Scaling Generally, we are interested in similarity indexes that are invariant to isotropic but not necessarily non-isotropic scaling

representational similarity matrices: pairwise similarity then compare similarity of structures. using an inner product of similarity, the similarity between matrices reduces to another intuitive notion of pairwise feature similarity. They have a nice summary of all the similarity metrics,

- * Dot Product-Based Similarity.
- * Hilbert-Schmidt Independence Criterion
- * Centered Kernel Alignment: HSIC made invariant to isotropic scaling
- * Kernel Selection *In practice, we find that RBF and linear kernels give similar results across most experiments, so we use linear CKA unless otherwise specified.*

They also look at

- * linear regression: total fraction of variance explained by the fit (not really used for similarity)
- * CCA: find a basis that maximises the correlation
- * SVCCA- address sensitivity of CCA
- * projection weight CCA- address sens of CCA (normalized)
- * Neuron Alignment Procedures
- * Mutual Information

TODO: They did some maths: Linear CKA vs CCA & Regression

They did some experiments

- * A sanity check for similarity indexes: training two identical neural networks, with only difference random initialization, a similarity score should see layer 1 A = Layer 1 B. The wrong layers were selected as the most similar aka similarity metric was poor. CKA was the best, the other metrics (Liner CKA, CCA, SVCCA, PWCCA, lin regression) confused the layers. They made the network deeper, the layers start looking identical (the network sopts learning) **TODO: does it?**- pathological depth. They also tested it on ResNet which has skip connections. CKA was able to determine the structure of the network by comparing the layer representaitons. They also did a width experiment: as the width increases the CKA tends towards 1, earlier layers converge first. **TODO: bias?**

Representations learned by independently trained networks often share measurable global geometric structure. CKA holds some geometric information.

Assumtions The same stimuli are presented to both models. Pairwise geometric relationships among stimuli are the relevant object. Orthogonal invariance is desirable, but invariance to arbitrary invertible transformations is not. The layer-correspondence test is a reasonable validation criterion. A global similarity statistic can be interpreted without a carefully specified null model.

The high-dimensional issue later emphasized by Gröger et al. is particularly important: a nonzero CKA score under independence is not automatically evidence of learned correspondence if the representation dimension is large relative to the number of examples.

2.1.6 SUPPORTING OR CONTRADICTING SOURCES

A pair of models can agree at level 4 but disagree at level 3. They can agree at level 1 while disagreeing at levels 3–5. Or they can have high CKA while differing functionally on out-of-distribution inputs. This is supported by model-stitching work. <chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://proceedings.neurips.cc/paper/2021/file/2cb274e6ce940f47beb8011d8ecb1462-Paper.pdf>

It finds that CKA, CCA, and Procrustes can perform differently depending on the functional criterion, with no single metric universally winning. (<https://bair.berkeley.edu/blog/2021/11/08/similarity/>)

2.1.7 WAVE2VEC [1]

self-supervised learning of representations from raw audio data CNN -> mask -> transformer **TODO: Gumle softmax** Contrastive Loss.

2.1.8 HUBERT [5]

A blog about how BERT wave2vec and HuBERT works: <https://thegradient.pub/an-illustrated-tour-of-applying-bert-to-speech-data/>

Uses a cluster-method. (k-means), use Pseudo-Labeling. Different signals differ from text and images that they are continuous-valued sequences.

Hidden unit BERT (HuBERT) that benefits from an offline clustering step to generate noisy labels for a BERT-like pre-training.

The predictive loss is only applied over the masked regions, forcing the model to learn good high-level representations of unmasked inputs to infer the targets of masked ones correctly

it either matches or improves upon the state-of-the-art wav2vec 2.0

They use the k-means clustering within the algorithm. I wonder how this would impact measuring the similarity, since it already optimises for this objective.

2.1.9 DEEP MULTIMODAL SEMANTIC EMBEDDINGS FOR SPEECH AND IMAGES [3]

present a model which takes as input a corpus of images with relevant spoken captions and finds a correspondence between the two modalities.

CNN (speech) CNN image, and an alignment and embedding model. With a fixed output mapped to an embedding space.

it was originally inspired by the text-to-image alignment models presented by Karpathy in [6, 8]. In [6], Karpathy uses a refined version of the alignment model presented in [8] to produce training exemplars for a caption-generating RNN language model that can be conditioned on visual features. Through the alignment process, a semantic embedding space containing both images and words is learned

In its simplest sense, our model is designed to calculate a similarity score for any given image and caption pair, where the score should be high if the caption is relevant to the image and low otherwise. By taking the dot product of an image embedding vector with an audio caption embedding vector, we obtain a similarity score S .

To further examine the high-level acoustic representations learned by our networks, we extracted spectrograms for 1645 instances of 14 different ground truth words from the development set by force aligning the Google recognizer hypotheses to the audio. We did a forward pass of each of these individual words through the audio branch of our network, leaving us with an embedding vector for each spoken word instance. We performed t-SNE [32] analysis on these points, shown in Figure 5. We observed that the points form pure clusters, indicating that the top-level activations of the audio network carry information which is discriminative across different words.

2.1.10 EXTRACTING SENTENCE EMBEDDINGS FROM PRETRAINED TRANSFORMER MODELS [11]

They explain the representations of transformer embedding models (often BERT). A key point mentioned about outliers in embeddings spaces was that removing outliers: "(citation) showed that this brings improvements for tasks that directly use the geometry of the embeddings", this i also read in one of the other papers, i think the Plato's Cave paper mentioned this. For the purpose of representation learning, removing the outliers for geomeryt metrics is a good idea.

This paper also mentions isotropy and the effect of the frequency of words in a training corpus resulting in the different representaitons, but since i will be lookinh at text image and audio clusters and similarity of representations, this might not be as relevant. They do consider the idea of the "narrow cone" in the embedding space. They also mention that early layers learn more general features while the later layer (especially the last layers) become very specific to the tast they are trained for.

They also mention various approaches and attempts to find the best representation of a sentence embedding and found that average pooling just works the best (this is also what the Aritotelian and cave paper did)

METHOD

3.1 MODELS

The transformer models used in previous research [2, 6, 7] include, **Text:** Platonic: 17 ViT models, BLOOM, OpenLLaMa, LLaMa, OLMo, LLaMa3, Gemma, Mistral/Mixtral BLOOM/BLOOMz - instruction tuned, Autoregressive OpenLLAMA, LLaMa3, OLMo, Gemma, Mistral, Mixtral, - Autoregressive (decoder-only), DeepSeek, Qwen, Falcon3, Yi-1.5, Granite, GPT-OSS,

Image: *How to transform the images to feed to the models* ImageNet ViT, MAE, DINO DINOv2, CLIP ViT, 17 ViT models ranging from ViT-tiny to ViT-giant, (trained on different tasks),

This means they do not force every image model to use one manually chosen resize and normalisation scheme. Instead, each model receives the preprocessing specified by its own pretrained configuration

Audio:

Speech Encoders: ASR: vq-wav2vec, wav2vec 2.0, Mockingjay, , Conformer, DeCoAR2.0, w2v-Conformer, W2v-BERT, HuBERT, XLS-R, UniSpeech UniSpeech-SAT, BigSSL, WavLM, SpeechT5, data2vec, data2vec2, SpeechFormer, Whisper

Leaving XLS-R by facebook since its optimized for multilingual based on wave2vec and this dataset is primarily in english and the objective is to test mutlimodal representations not multi-lingual. Include the wave2vec robust modles, pretrained on multiple domains. arxiv:2104.01027 WavLM, , encoder, Tacotron, Tacotron 2, Transformer TTS

The main differences between wav2vec and w2v-BERT lie in their pre-training objectives and the data types they operate on. [9]

Multi-modal transformers? data2vec, deltaLM, SpeechT5, SpeechFormer However, unlike wav2vec, which focuses solely on speech signals, data2vec can operate on various types of data and can learn joint representations that capture crossmodal correlations and transfer knowledge across modalities.

Some of them have different training objectives, but the performance are not being measured, only simialrity.

The model selection allows us to ask and investigate mutiple questions. By using constrolled subsets. Inner family scale is comparable. (LLama, HubERT, DINO, wisper)-for Platonic experiment

Convergence over different training objective: MAE, DINOv2, CLIP, AugRegVIT (all rouhgly VIT based but different objectives)

wave2vec, HuBert, data2vec, WavLM.

Despite being trained using different objectives, do strong models recover similar representations?

Does the same training framework lead to cross-modal convergence? data2vec-text, audio (note not speech), vision

Table 3.1: Language models selected for representational similarity analysis. Models within the same family provide controlled comparisons across scale, while the complete set provides variation in model family and training objective.

Family	Models / scales	Architecture	Training
BLOOM	560M, 1.1B, 1.7B, 3B, 7.1B	Decoder Transformer	Causal language modelling
BLOOMZ	560M, 1.1B, 1.7B, 3B, 7.1B	Decoder Transformer	BLOOM models with multilingual multitask instruction tuning
OpenLLaMA	3B, 7B, 13B	Decoder Transformer	Causal language modelling
LLaMA	7B, 13B, 30B, 65B	Decoder Transformer	Causal language modelling
Llama 3	8B, 70B	Decoder Transformer	Causal language modelling
Gemma	2B, 7B	Decoder Transformer	Causal language modelling
Mistral	7B	Decoder Transformer	Causal language modelling
Mixtral	8×7B	Sparse MoE Transformer	Causal language modelling
OLMo	1B, 7B	Decoder Transformer	Causal language modelling
data2vec Text	Base	Transformer encoder	Self-supervised latent representation prediction

How these model groups support the experiments

The selected models enable several complementary comparisons. First, models of different sizes within the same family provide relatively controlled tests of whether representational alignment increases with model scale. Examples include the BLOOM, LLaMA,

Table 3.2: Vision models selected for representational similarity analysis. The model set contains several ViT families trained using substantially different learning objectives, allowing convergence to be evaluated across both scale and training method.

Family	Models / scales	Architecture	Training
AugReg ImageNet-21K	Tiny, Small, Base, Large	Vision Transformer	Supervised image classification / augmentation and regularisation
MAE	Base, Large, Huge	Vision Transformer	Self-supervised masked image reconstruction
DINOv2	Small, Base, Large, Giant	Vision Transformer	Self-supervised representation learning / self-distillation
CLIP	Base, Large, Huge	Vision Transformer	Contrastive image-text learning on LAION-2B
CLIP + fine-tuning	Base, Large, Huge	Vision Transformer	Contrastive image-text pretraining followed by image classification fine-tuning
data2vec Vision	Base, Large	Vision Transformer	Self-supervised latent representation prediction

DINOv2, HuBERT, and Whisper families.

Second, models within the same modality but trained using different objectives allow the robustness of representational convergence to training methodology to be investigated. In vision, this includes supervised ImageNet models, MAE, DINOv2, and CLIP. In speech, this includes Wav2Vec 2.0, HuBERT, data2vec, WavLM, W2v-BERT, and Whisper.

Third, the data2vec models provide a useful cross-modal controlled comparison. data2vec applies the same general self-supervised representation-learning framework independently to text, vision, and speech. Alignment between the three data2vec representations can therefore be compared with alignment between models that differ in both modality and learning objective.

Finally, W2v-BERT and Whisper provide architectural and training diversity within the speech model set. W2v-BERT uses a Conformer architecture, combining convolution with self-attention, while Whisper uses an encoder-decoder Transformer trained with speech-text supervision. These models can therefore test whether observed representational convergence generalises beyond conventional self-supervised Transformer speech encoders.

Model sizes:

- * DINOv2: small, base, large, giant
- * wav2vec: base, large, largeLV60
- * Hubert: base, large, Xlarge

Table 3.3: Speech models selected for representational similarity analysis. The majority are self-supervised speech encoders, complemented by W2v-BERT as a Conformer-based model and Whisper as a supervised encoder–decoder family. Only Whisper encoder representations are used in the similarity analysis.

Family	Models scales	/	Architecture	Training
Wav2Vec 2.0	Base, Large		CNN + Transformer encoder	Self-supervised contrastive speech represen- tation learning
Wav2Vec 2.0 variants	Large-Robust, Large-LV60		CNN + Transformer encoder	Self-supervised; similar model scale with differ- ent pretraining data
HuBERT	Base, Large, XLarge		CNN + Transformer encoder	Self-supervised masked pre- diction using clustered speech targets
data2vec Audio	Base, Large		CNN + Transformer encoder	Self-supervised latent rep- resentation prediction
WavLM	Base, Base+, Large		Transformer-based speech encoder	Self-supervised speech represen- tation learning
W2v-BERT 2.0	2.0		Conformer	Self-supervised contrastive and masked speech modelling
Whisper	Tiny, Base, Small, Medium, Large		Encoder–decoder Transformer	Large- scale super- vised/weakly supervised speech recogni- tion; encoder only used

* WavLM: Base, base+, large

3.2 EXPERIMENTS

The primary goal of the experiments is to investigate whether independently trained transformer models converge towards shared representations, and whether any observed convergence is global, local, or modality-specific. The study will first reproduce the central experiments of the Platonic Representation Hypothesis on a new paired dataset, using model families similar to those evaluated by Huh et al. [6]. Where computationally feasible, the model set will be extended to include newer models considered by Koepke et al. [7].

The second goal is to evaluate the Aristotelian Representation Hypothesis, which proposes that increasingly capable models converge primarily in their local neighbourhood structure rather than towards an identical global geometry. To distinguish these hypotheses, both global and local representational similarity metrics will be evaluated.

The final goal is to extend the image–language analysis to speech. This will determine whether speech representations participate in the same cross-modal convergence observed between vision and language, and whether any convergence is better characterised as global geometric agreement or local topological agreement.

The experiments aim to address the following research questions specific to the Flickr8k dataset,

1. Reproduce the image–language experiments of Huh et al. [6] on a new paired dataset.
2. Extend the same experimental protocol to include speech representations.
3. Evaluate whether the observed convergence is better explained by the Platonic Representation Hypothesis or the Aristotelian Representation Hypothesis.

3.2.1 CORE EXPERIEMENTS

Experiment 0: Extract the layer embeddings from models

"For the vision model, we use class token of each layer, and for the language model, we average pool each layer to a single token"[6] "compute pairwise alignment scores, then take the maximum One of these pairwise comparisons also includes concatenated features. We apply l2 normalization to the features before measuring the distance. As transformer architectures have “emergent outliers” (Dettmers et al., 2022), we truncate the elements in the features that are above the 95-th percentile."

For every model, representations will be extracted at multiple layers. Following Huh et al. [6], the class token or pooled representation will be used for vision transformers, while mean pooling over valid tokens will be used for language models. **TODO: Speech representations will be pooled over valid acoustic frames, excluding padded positions.**

Where a model provides a dedicated pooled embedding, this will be evaluated separately from mean pooling.

A layer-wise similarity matrix can be computed and a maximum layer-pair score reported to reproduce Hug [6]. However, because maximising over a larger number of layer pairs introduces a depth-related selection bias, aggregation-aware null calibration will also be applied. *Therefore both raw and calibrated results will be reported.*

Multiple similarity metrics will be used to measure different types of similarity in the embedding space. Two types of similarity will be investigated.

- * *Global similarity*: linear and RBF Centered Kernel Alignment, with additional alignment metrics: linear CKA, RBF CKA, unbiased CKA, SVCCA, PWCCA, RSA, **TODO: RV coefficient**, orthogonal Procrustes similarity or distance.
- * *Local similarity*: mutual-kNN, cycle kNN, CKNNA,

These metrics (Global) test whether two representations share a similar global geometry, covariance structure, or pairwise distance structure. These measures (Local) test whether two models agree on which samples are neighbours, without requiring agreement on exact pairwise distances.

Experiment 1.1: image-text convergence The second experiment will reproduce the central image–language analysis of the Platonic Representation Hypothesis on the new dataset. A set of vision models and language models will be evaluated at *multiple scales*.—maybe scales. Candidate language models should include multiple scales within the same family to separate model-family effects from scaling effects.

For every vision–language model pair, representations of aligned image–text samples will be compared. Layer-wise similarities will be computed, followed by both the raw and calibrated aggregate score.

The following comparisons will be reported:

1. raw linear and RBF CKA;
2. calibrated linear and RBF CKA;
3. raw and calibrated mKNN;
4. cycle- k NN and CKNNA;
5. selected geometric metrics such as SVCCA, RSA, and Procrustes similarity.

The Platonic hypothesis predicts that both global and local alignment increase with capability. The Aristotelian hypothesis predicts that the scaling trend will be substantially weaker for calibrated global metrics, while remaining visible for local neighbourhood metrics.

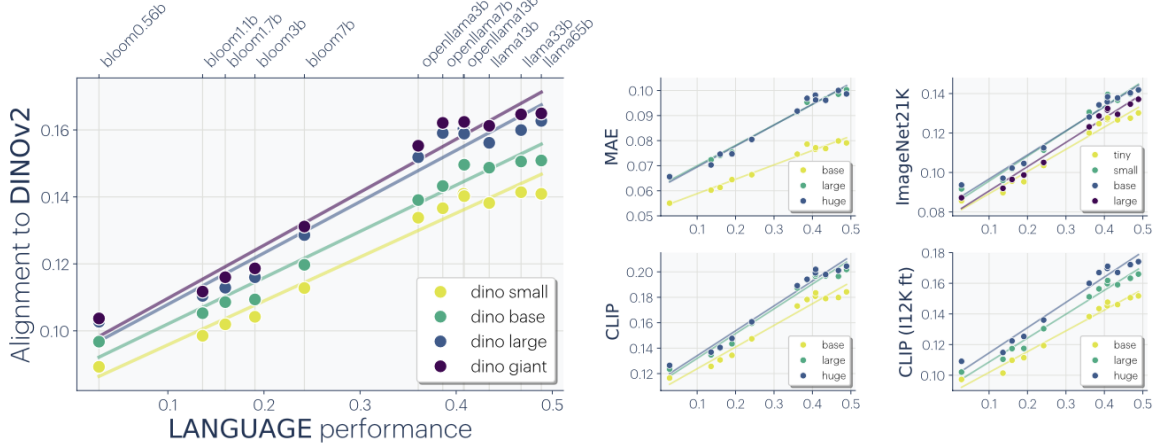


Figure 3. LANGUAGE and VISION models align: We measure alignment using mutual nearest-neighbor on the Wikipedia caption dataset (WIT) (Srinivasan et al., 2021). The x-axis is the language model performance measured over 4M tokens from the OpenWebText dataset (Gokaslan & Cohen, 2019) (see Appendix B for plots with model names). We measure performance using 1 – bits-per-byte, where bits-per-byte normalizes the cross-entropy by the total bytes in the input text string. The results show a linear relationship between language-vision alignment and language modeling score, where a general trend is that more capable language models align better with more capable vision models. We find that CLIP models, which are trained with explicit language supervision, exhibit a higher level of alignment. However, this alignment decreases after being fine-tuned on ImageNet classification (labeled CLIP (I12K ft)).

Figure 3.1: Figure 3 from [6]

We follow the experimental protocol of Huh et al. (2024) using $n = 1024$ image-text pairs (WIT; Srinivasan et al. (2021)) and embeddings from three language model families (BLOOM, OpenLLaMA, LLaMA) and five vision model families (ImageNet-21K, MAE, DINOv2, CLIP, CLIP-finetuned) across multiple scales. As in Huh et al. (2024), we extract per-layer representations using the class token for vision models and mean pooling over tokens for language models.

For each pair, we compute layer-wise similarity and report the maximum across layers, as in the original work

We evaluate both global spectral metrics (CKA linear/RBF) and local neighborhood metrics (mKNN, cycle-kNN, CKNNA).

global similarity: measure CKA and also the calibrated metric local similarity.

I don't have super different architectures- only transformers We expand upon this observation by evaluating the transfer performance of 78 vision models. These models were trained with varying architectures, training objectives, and datasets (detailed in Appendix C.1). In Figure 2 (left), we bin these models based on their average transfer performance on the VTAB dataset (Zhai et al., 2019), and then measure the average kernel alignment of the models within each bin. The results indicate that models with high transfer performance form a tightly clustered set of representations, while models with weak performance have more variable representations. We further visualize this structure with UMAP (McInnes et al., 2018) over models representation in Figure 2 (right). This suggests that models that are competent all represent data in a similar way. Echoing Bansal et al. (2021) and Tolstoy (1877), we might say: all strong models are alike, each weak model is weak in its own way.

They investigate the alignment between language and vision, this may be extended to lang-speech and visioin-speech.

Experiment 2.1: speech-text convergence The third experiment will evaluate whether speech and language models learn aligned representations. The same metrics used in the image-language experiment will be applied.

The analysis will determine whether more capable language models align more strongly with speech representations, and whether stronger speech encoders produce greater alignment across language-model families.

A critical consideration is that speech contains information that is absent from a transcript, including speaker identity, pitch, prosody, accent, background conditions, and timing. Therefore, speech-language alignment is not expected to become perfect. The experiment instead tests whether the modalities recover a shared semantic or linguistic neighbourhood structure.

Experiment 2.2: Image-speech convergence The fourth experiment will compare image and speech representations directly. Image-speech alignment is likely to be weaker than image-text or speech-text alignment because the two modalities may share less information directly. Nevertheless, significant local alignment would support the claim that both modalities encode shared semantic relationships.

Experiment 2.3: Tri-modal consistency - maybe

Once pairwise image-text, speech-text, and image-speech scores are available, the consistency of alignment across all three modalities can be investigated. A tri-modal neighbourhood overlap score may then be defined. This provides a direct test of whether all three modalities identify the same semantic neighbours. It should be treated as an exploratory extension because it is not part of the original Platonic or Aristotelian protocols.

Experiment 4: Dataset-size scaling

The available dataset contains approximately 8,000 aligned samples, compared with the approximately 1,024 samples used in the original WIT experiments. This enables a controlled analysis of sample-size sensitivity.

Nested subsets of size

$$n \in 1000, 2000, 4000, 8000$$

3.2.2 ANALYSES

Analysis : Neighbourhood-size and locality analysis The Aristotelian hypothesis specifically concerns local neighbourhood structure. The choice of neighbourhood size must therefore be examined.

For mutual nearest-neighbour metrics, evaluate

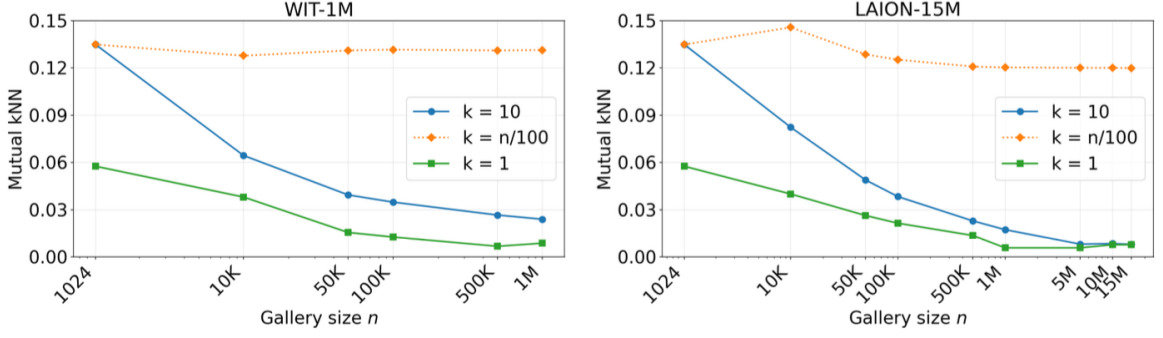


Figure 4: Scaling the gallery size to 1M (WIT) and 15M (LAION) shows a large drop in mutual k NN alignment for $k=1$ and $k=10$ for DINOv2 and OpenLlama features.

Figure 3.2: Figure 4 from [7]

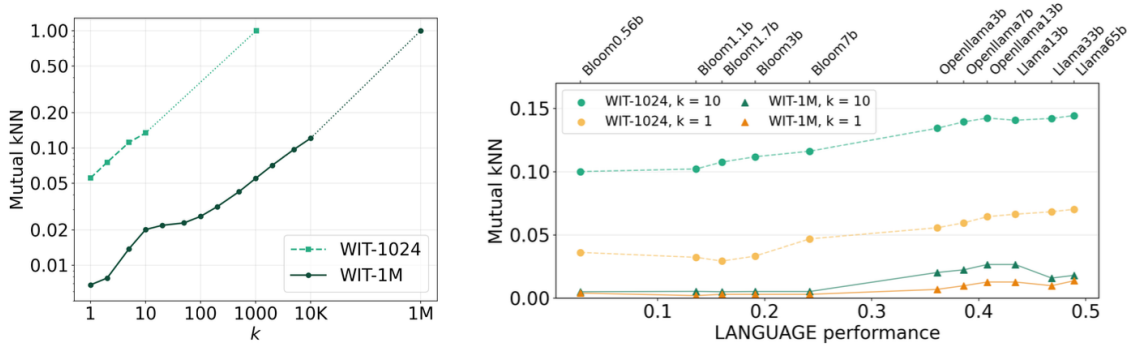
$$k \in 5, 10, 20, 50, 100,$$

or choose comparable values relative to dataset size. A proportional alternative is

$$k \in \{0.1\%, 0.5\%, 1\%, 5\%, 10\%\} \times n.$$

They investigate the effect of the size of K across the models.

In Figure 10, we compute the average alignment score across all LLM models. For each k , we center the scores to the smallest vision model and divide by the standard deviation



(a) Effect of neighborhood sizes k in mutual k NN (both axes log-scaled). Trivially, mutual k NN converges to 1.0 as k approaches the full gallery size. [42] utilized mutual k NN for $k=10$.

(b) Alignment between DINOv2 and different LLMs, measured on WIT-1024 and WIT-1M. As observed in [42], alignment (mutual k NN) increases with language performance (measured as 1 – bitsperbyte from [42]) on the 1024-sample set, but this trend breaks with larger gallery size.

Figure 3: Mutual k NN text-image feature alignment when scaling from WIT-1024 to WIT-1M. (a) shows the dependence on neighborhood size k , while (b) examines alignment for different LLMs. The observation from [42], that more capable language models align better with vision at fixed small k largely vanishes at WIT-1M scale.

Figure 3.3: Figure 3 from [7]

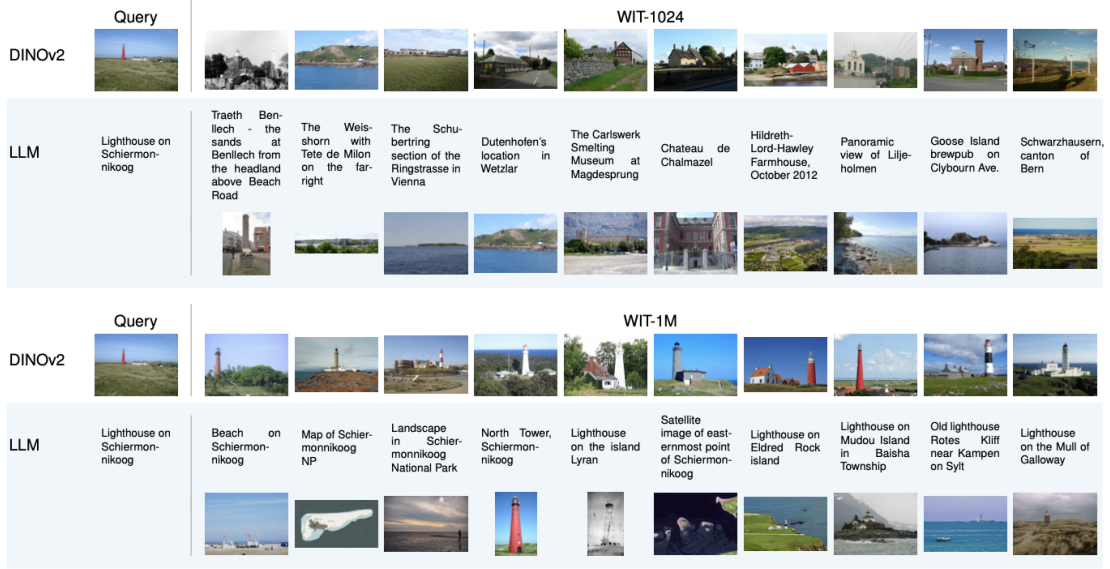


Figure 2: **Nearest-neighbor quality depends on data density.** We show 10 within-modality nearest neighbors for image (DINOv2) and text (LLM) embeddings on a sparse WIT-1024 gallery (top) and a denser WIT-1M gallery (bottom). For text queries, retrieved captions and their corresponding reference images are shown. At smaller scale, nearest neighbors are less semantically precise. Nearest-neighbor structure becomes more semantically refined as gallery density increases.

Figure 3.4: Figure 2 from [7]

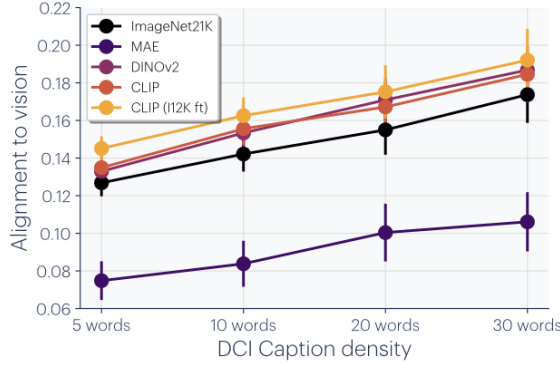


Figure 9. **Increasing caption density improves alignment:** We vary caption length using the Densely-Captioned-Images (DCI) dataset (Urbanek et al., 2023). Starting from a dense caption, we used LLaMA3-8B-Instruct (Meta, 2024) to summarize and generate coarse-grained captions. We compute the average alignment score across all vision and language models with standard deviation measured over the language models we evaluated. With denser captions, the mapping may become more bijective, leading to improved language-vision alignment scores.

Figure 3.5: Figure 9 from [6]

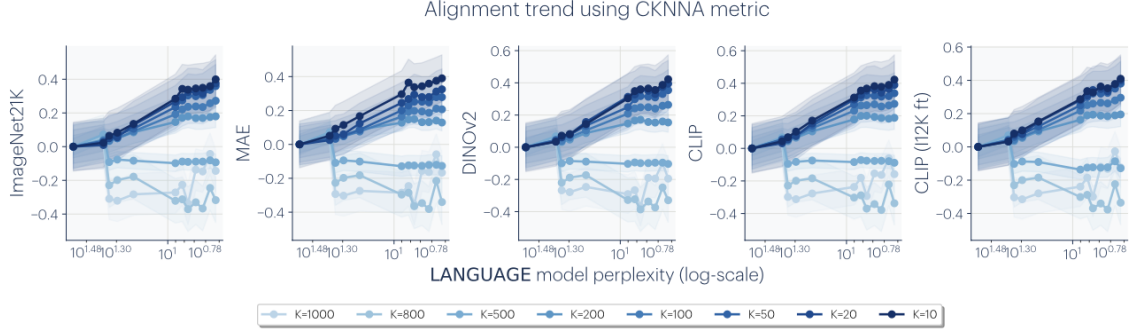


Figure 10. Cross-modal alignment increases locally: Alignment trend when varying the top- k nearest neighbors in the CKNN metrics (Eqn. 18). We center alignment score to the smallest language model and divide the total trend by the standard deviation. When $k = 1024$, we recover the original CKA metric, and when $k < |\mathcal{X}|$ it closely resembles the mutual nearest-neighbor metric m_{NN} . Each line represents the average of all LLM models for a specific k . As we decrease k , the alignment becomes more pronounced.

Figure 3.6: Figure 10 from [6]

of the scores. We find that high values of k show less conclusive alignment across tasks while decreasing k shows a coherent trend across both models and tasks.

Analysis: Layer-selection sanity checks

Analysis: Deduplication and contamination controls In our analyses at larger gallery scales, we perform deduplication to prevent near-duplicate samples from trivially inflating neighborhood overlap

Analysis: Nearest-neighbour retrieval analysis Quantitative alignment scores should be supplemented with qualitative nearest-neighbour queries.

For selected query samples, the top- k neighbours will be retrieved independently from image, text, and speech representation spaces. The analysis should report:

- neighbours shared across modalities;
- neighbours unique to one modality;
- semantic similarities;
- speaker or acoustic similarities;
- visually similar but semantically different examples; and
- semantically similar but acoustically or visually different examples.

Analysis: Speaker and acoustic-factor analysis

- speaker identity;
- accent;
- sex or voice characteristics, where ethically and appropriately available;
- recording environment;
- utterance duration;
- pitch statistics;
- speaking rate;
- signal-to-noise ratio; and
- semantic category.

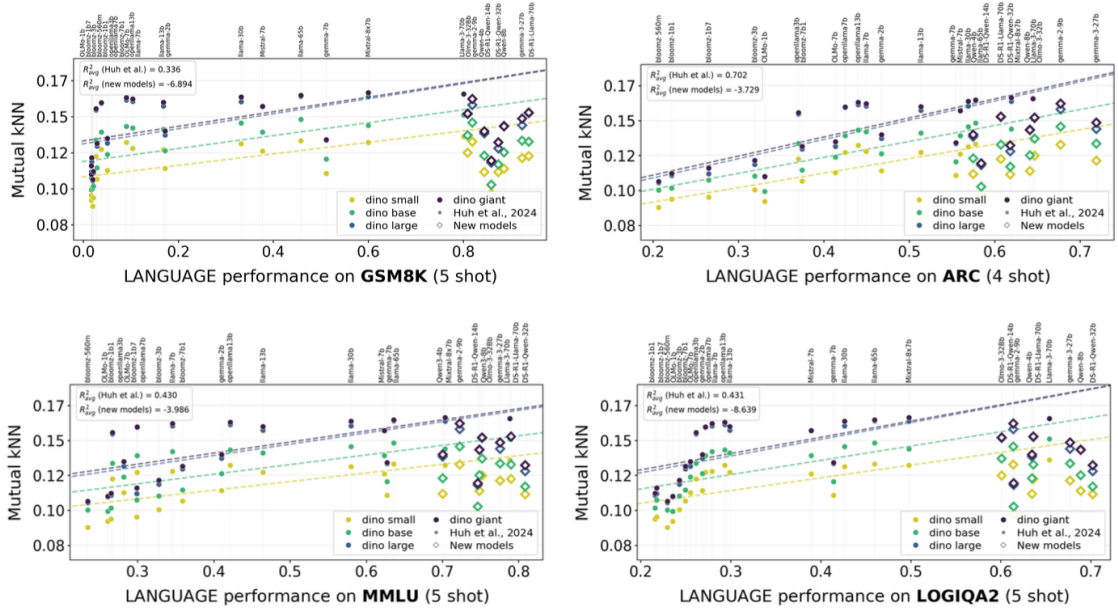


Figure 12: Testing whether the alignment-LLM performance trend from [42], tested on WIT-1024, holds for recent LLMs on ARC, GSM8K, MMLU, and LogiQA2. Dashed lines show the trend for models from [42] (circles). Recent LLMs (diamonds) do not follow the trend: stronger language models do not seem to be more aligned with DINOv2.

Figure 3.7: Figure 12 from [7]

Neighbourhoods can then be inspected to determine whether the nearest neighbours share semantic content, speaker identity, or acoustic properties.

Experiment x: Color test - perceptual organization

Experiment x: Within-modality convergence Similar to Figure 2 from [6], The first experiment will test whether stronger models within a single modality learn more similar representations. Models are grouped by modality and family. For each modality, pairwise representational alignment will be calculated between models trained using different.

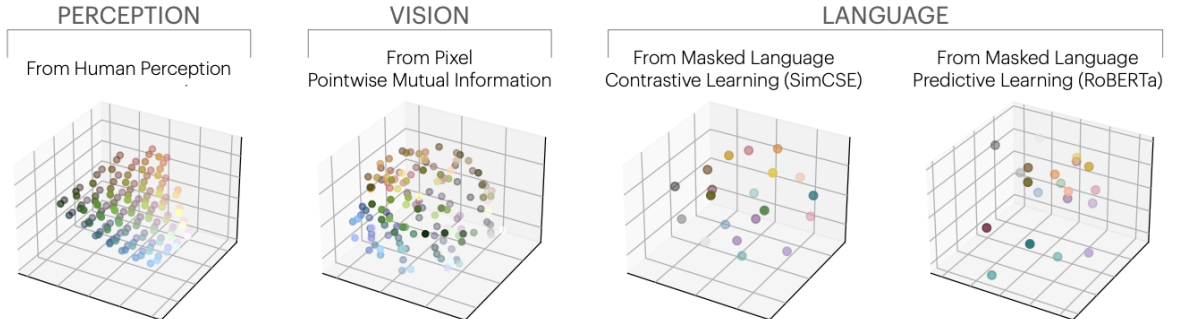


Figure 8. Color cooccurrence in VISION and LANGUAGE yields perceptual organization: Similar representations of color are obtained via, from LEFT to RIGHT, the perceptual layout from CIELAB color space, cooccurrence in CIFAR-10 images, and language cooccurrence modeling (Gao et al. (2021); Liu et al. (2019); computed roughly following Abdou et al. (2021)). Details in Appendix D.

Figure 3.8: Figure 8 from [6]

The experiment answers the hypothesis:

More capable models within a modality exhibit greater representational alignment than weaker models.

Report: raw and calibrated alignment

We sampled a variety of models trained either solely on vision or solely on language, and compared their representations as they became larger and more competent over many tasks. In Figure 3, we assess alignment between a suite of language models and vision models. So far we have only defined alignment for two kernels defined over the same input space. To measure cross-modal alignment, we use paired datasets to bridge the two modalities

SUMMARY AND CONCLUSIONS

REFERENCES

- [1] Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [2] Fabian Gröger, Sheng Wen, and Maria Brbić. Revisiting the platonic representation hypothesis: An aristotelian view. *arXiv preprint arXiv:2602.14486*, 2026.
- [3] David Harwath and James Glass. Deep multimodal semantic embeddings for speech and images. In *IEEE Workshop on Automatic Speech Recognition and Understanding (ASRU)*, 2015.
- [4] David Harwath, Antonio Torralba, and James Glass. Unsupervised learning of spoken language with visual context. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- [5] Wei-Ning Hsu, Benjamin Bolte, Yung-Hsuan H. Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed. Hubert: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29:3451–3460, 2021.
- [6] Minyoung Huh, Brian Cheung, Tony Wang, and Phillip Isola. Position: The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [7] A. Sophia Koepke, Daniil Zverev, Shiry Ginosar, and Alexei A. Efros. Back into plato’s cave: Examining the evidence for cross-modal representational convergence. *arXiv preprint arXiv:2604.18572*, 2026.
- [8] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 3519–3529, 2019.
- [9] Siddique Latif, Aun Zaidi, Heriberto Cuayáhuitl, Fahad Shamshad, Moazzam Shoukat, and Junaid Qadir. Transformers in speech processing: A survey. *arXiv preprint*

-
- arXiv:2303.11607*, 2023. doi: 10.48550/arXiv.2303.11607. URL <https://arxiv.org/abs/2303.11607>.
- [10] Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. Svcca: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. *arXiv preprint arXiv:1706.05806*, 2017. URL <https://arxiv.org/abs/1706.05806>.
- [11] Lukas Stankevičius and Mantas Lukoševičius. Extracting sentence embeddings from pretrained transformer models. *arXiv preprint arXiv:2408.08073*, 2025. doi: 10.48550/arXiv.2408.08073. URL <https://arxiv.org/abs/2408.08073>.
- [12] Hao-Ting Wang, Jonathan Smallwood, Janaina Mourão-Miranda, Cedric Huchuan Xia, Theodore D. Satterthwaite, Danielle S. Bassett, and Danilo Bzdok. Finding the needle in a high-dimensional haystack: A tutorial on canonical correlation analysis. *CoRR*, abs/1812.02598, 2018. URL <https://arxiv.org/abs/1812.02598>.

APPENDIX A

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